Concrete finishing: a pump pad, joints and a patch

Gym-Anything for Robotics, environment `concrete-finishing`.

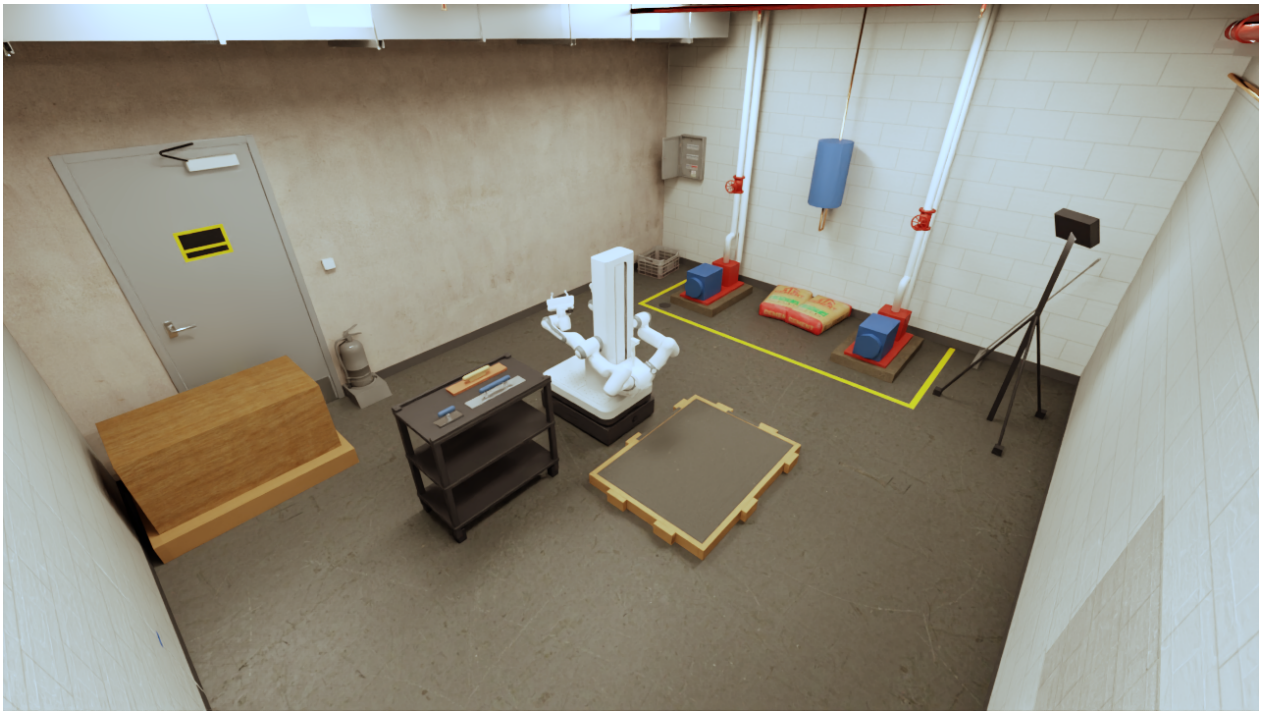


Figure 1. Isaac view of mechanical room B-012. The routine policy camera is a software renderer of the same scene (section 7).

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0 Summary

The environment simulates finishing five pastes by hand in a mechanical room: concrete on a new pump pad, repair mortar in a floor patch, mortar joints in a block infill, sealant in a control joint and compound on a wallboard patch. The paste model has conservative material packets and spatial catches, a finite surface-paste reserve, age-aware skin mechanics, a fibre contact envelope with a compliant knife sole, and a confined-work surface consolidation of mortar with strength feedback. The routine policy camera is a CPU renderer of the same scene geometry, with finish optics that normalize unresolved relief, and an actual wrist-camera view is shown.

Component checks show local darby, edger and T-junction finishing, a positive and a negative local joint comparison, six-pass smooth-hard troweling and the recovery of one mortar tag, with compact recordings and explicit failure examples. Remaining limits include incomplete joint coverage and cleanup, coarse mesh and contact approximations and incomplete calibration of response laws (section 9). The humanoid family is not supported. The environment does not come with a nine-task autonomous solution.

1 The activity and why this scenario

The action group is **smooth cement edge**: 39 occupational task statements (O*NET 30.3) from 26 occupations, merged into one canonical action by the grounding pipeline, with an allocated value of 7.11 billion USD a year and no personal-time share. The rows are not only concrete flatwork. Grouped by what is being smoothed:

Kind of work	Rows	Source weight	Occupations
Fresh concrete: float, trowel, edge	12	8.44 B USD	construction laborers, cement masons, carpenters' helpers, roustabouts, landscaping workers, millwrights
Patch, fill or repair flush (cementitious)	8	3.03 B USD	cement masons, terrazzo workers, mason helpers, foundry and molding workers
Wall filler, putty, joint compound	8	3.92 B USD	painters, tapers, glaziers, weatherization installers
Sealant or caulk bead tooling	4	5.09 B USD	construction laborers (caulking), painters, drywall installers, auto glass installers
Masonry and tile joints	5	1.35 B USD	brickmasons, stonemasons, tile setters, mason helpers
Other (pottery seam, dental acrylic)	2	0.20 B USD	potters, dental laboratory technicians

What the rows share is one physical act: a hand-held rigid tool is pressed onto a paste that yields above a stress and stiffens with time, and slid, pitched and timed so that the paste ends flat, flush, concave or sealed. The tool and the paste are different in each row (a magnesium float on concrete, a jointer in mortar, a spatula in a sealant bead, a knife on joint compound), and so is the clock: concrete must be floated after its bleed water is gone and before it sets, mortar tooled when "thumbprint hard", a sealant tooled before it skins.

The scenario puts five of these pastes in one ordinary place, the way they occur on a real job: a **mechanical room during a pump replacement**. The crew has just cast a new housekeeping pad for the pump (1219 × 914 mm in 2×4 forms on the existing slab), patched the floor where the old pad was broken out, filled an old duct opening in the north CMU wall with block, gunned sealant into the east wall's control joint, and spackled the anchor holes of removed conduit straps in the gypsum partition. Every one must now be finished by hand at the right moment. Nine tasks cover the pad's four stages and the whole pad, the patch, the joints, the sealant and the wallboard. The research weighed four other settings (a sidewalk panel replacement, a bathroom retile, a precast yard, a façade repair) and chose this one because it holds the most of the group's rows at full fidelity in one robot-reachable room.

2 The scene

2.1 Room B-012

A pump room in a basement, 6.10 × 4.88 m with the deck at 3.05 m (frame: x east, y north, z up, origin on the floor in the south-west corner). The north, east and west walls are 8 in concrete masonry units (actual $7\frac{5}{8} \times 7\frac{5}{8} \times 15\frac{5}{8}$ in, 3/8 in joints), painted, with concave tooled joints; the south wall is a gypsum partition with a hollow metal door. The floor is the building's slab, steel-troweled and worn, with a drain and painted safety lines. Two old base-mounted pumps stand on their pads on the west side (their tooled edges show what a finished pad looks like), the new pump waits crated on its pallet by the door, and copper, EMT conduit, a duct, an expansion tank, cement bags, a broom, a sign and an extinguisher dress the room. Four 2×4 ft troffers (4000 K) light it to about 350 lx at the floor, and an LED work light on a stand rakes across the pad, as finishers light a slab to see its sheen.

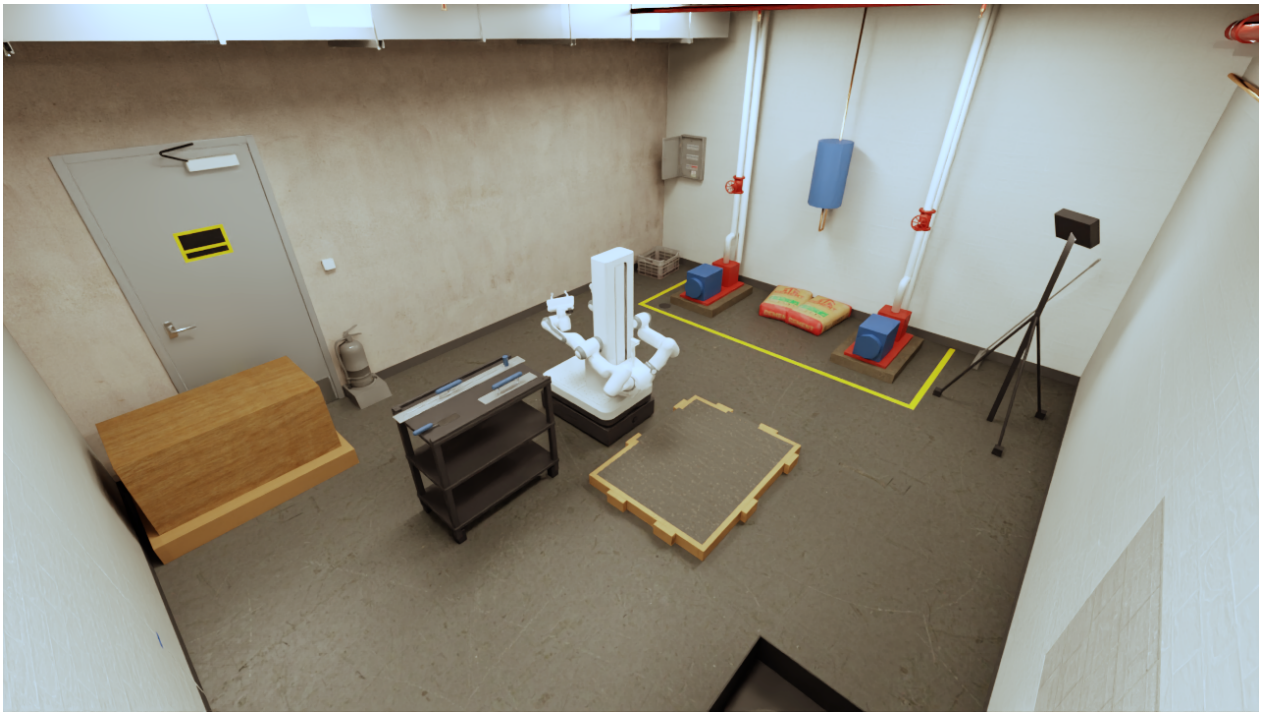


Figure 2. The room from its north-east corner (the `room` view), at the start of `pad-darby`.

2.2 The stations and what is on them

Station	Where	What it is
Pad	centre (3.20, 2.90) m	1219 × 914 mm inside 2×4 forms (38 × 89 mm dressed) on edge, braced by kickers every 0.6 m; 89 mm of concrete
Floor patch	(4.60, 1.25) m	a saw-cut recess 400 × 250 mm, chipped 25–38 mm deep, filled with repair mortar
Block infill	north wall, x 3.89–5.11 m, sill at 0.81 m	four courses of 8 in block, three to a course in running bond, 3/8 in joints
Control joint	east wall at y 2.00 m	13 mm wide, backer rod 8.5 mm behind the face, the task segment z 0.45–2.00 m between two strips of tape
Wallboard	south partition, around (1.80, 1.25) m	a 38 mm hole closed with a mesh patch and spackled; four 6 mm anchor holes

Each station has its tool cart (0.86 × 0.46 m, top at 0.84 m, casters locked) with the tools of the job lying on it as a crew leaves them: the darby, floats, trowels, edger, margin trowel and the pocket penetrometer at the pad; the floats and margin trowel at the patch; the jointer and a margin trowel with a mortar board on its stand at the infill; the sealant tool and a rag at the control joint; the joint and putty knives and the hawk with compound at the wallboard. A mixing tub with leftover concrete stands north of the pad for the darby task (the material for low spots).

2.3 The tools

Every tool is a rigid body with the maker's dimensions and an estimated mass (the shipping weight less 10–25 %); handles are round, as sold, not squared for the gripper. Their soles, the part that works the paste, are described to the paste model separately (section 3.6).

Tool	Part	Mass (kg)	Working sole (as the paste sees it)	Carries paste	Densifies	Seals
Magnesium hand float	145D	0.500	flat blade, 406 × 79 mm	no	0.3	0.1
Wood hand float	144D	0.280	flat blade, 406 × 89 mm	no	0.2	0.05
Finishing trowel	PB64D	0.450	flat blade, 356 × 102 mm	no	1	1
Magnesium darby	4626D	1.300	flat blade, 914 × 85 mm	no	0.2	0.05
Hand edger	136D	0.250	flat + quarter round + lip, 152 × 79 mm	no	0.6	0.2
Margin trowel	52D	0.120	flat blade, 127 × 51 mm	150 ml	0.5	0.3
Concave jointer	BJCC12	0.120	rod, 137 × 12 mm, r 6.3 mm	no	0.6	0
Sealant tool	5/8 in tooling spatula (Albion classic widths)	0.030	rounded nose, 18 × 16 mm, r 11.0 mm	15 ml	0.3	0
Joint knife	JK886D	0.160	flat blade, 152 × 70 mm	100 ml	0.5	0.3
Putty knife	PK878D	0.080	flat blade, 51 × 70 mm	25 ml	0.5	0.3
Pocket penetrometer	H-4134	0.227	flat blade, 6 × 6 mm	no	0	0

2.4 Start states

A task starts where a crew leaves the work, at the stage the task needs; nothing in the instruction states the age, which the robot must judge (by the sheen, the penetrometer, a fingertip, or the clock it is given).

Task	Material	Age at reset (21 °C)	What is there
`pad-darby`	concrete, slump 100 mm	5 min after strike-off	screed ridges, a low corner, a high band, coarse aggregate showing on 30 % of the surface, smear on two form tops, leftover concrete in the tub
`pad-edge`	same	60 min	darbied flat, a bleed film standing, square sharp edges against the forms
`pad-float`	same	150 min	darbied and edged (the edger's tracks pressed along the forms), the bleed water gone
`pad-trowel`	same	165 min	floated: sandy, open, float swirls
`pad-finish`	same	5 min	as `pad-darby`, the whole sequence to do
`patch-flush`	repair mortar	6 min after mixing	overfilled 3–7 mm, a void along the west edge, smear on the slab along two sides
`tool-joints`	Type N mortar	courses laid 45, 37, 28 and 20 min before	struck flush, crumbs on the block faces, one head joint 3 mm short
`tool-sealant`	one-part sealant	10 min	a bead 1.5–2.7 mm proud, two gun stops, a 40 mm stretch thin on one side
`patch-drywall`	wall compound	4 min	a mound 1–3 mm proud with knife ridges over the mesh, the anchor holes open, compound on the hawk

Seeds move the surface features; variants change the room's air (`door` open: a draught; a `fan` left running), the temperature (`hot` 27 °C, `cool` 15 °C, with the ages shifted so that each start is the same stage of the work) and the sealant grade (`fast-sealant`). The variants are a contract of the environment (`variations.json`); the instruction names the variant as a crew would see it ("The door to the corridor is open and there is a draught across the room.").



Figure 3. Start states, as the policy's head camera and the report's views see them: the pad before darbying, the patch, the infill joints, the sealant bead and the spackled wallboard.

3 Physical model

3.1 Engines, units and clocks

The rigid world is the repository's classroom world (Newton 1.5 with its MuJoCo solver on the processor, implicit fast integration, 0.5 ms ticks), reused unchanged: the Mobile FR3 Duo, its controllers and reflexes, the room's static colliders, the carts and every tool as a free body held by the robot's own fingertip pads through friction. The pastes are not rigid bodies. Each work surface is a **paste region**: a 2.5-dimensional field over a grid laid on the surface (1 to 3 mm cells), integrated by this environment's own code beside the rigid solver, and coupled to it through support plates (section 3.6). Everything is in SI units. There are two clocks: the rigid world's tick (0.5 ms) and the material clock, which advances with it while the robot works (material steps of 50 ms) and alone, in steps of 1 s, during a `wait` command (section 3.10).

3.2 What a cell holds

Per cell: the base height b (the rigid support under the paste: the slab, the recess floor, a block face, the backer rod, the board); the bulk thickness t_b and the surface layer t_s (the paste-rich top a float brings up); a blister lift; the bleed water film w and trapped water q; the share of coarse aggregate showing E; densification D and sealing S (what steel troweling does); the water-cement rise of the top 6 mm $\Delta w/c$ (water worked in); a texture amplitude a; the time it was placed t_0, sealed and last worked; for joints, contact with each joint wall and compaction; for every cell, whether it may hold paste and whether it is a finished reference face on which paste counts as smear. Thicknesses are kept in double precision (in single precision, ten thousand small deposits on an 89 mm depth lose 62 mm³ of a 99 L pad in 3.7 s).

3.3 Strength in time

The bulk yield stress grows with the material's age a:

$\tau_b(a) = \tau_b0 + A_thix \cdot a / s(T) + \tau_IS \cdot (a / (t_IS \cdot s(T)))\hat{}m$

For the pad's concrete, $\tau_b0 = 1566$ Pa is the yield stress of a 100 mm slump concrete (NISTIR 6094 eq. 19, opened); $A_thix = 0.25$ Pa/s is inside Roussel's ordinary thixotropy class (between 0.1 and 0.5 Pa/s, opened through Yu et al. 2025); the power law is anchored on the ASTM C403 set definitions (initial set at 3.45 MPa penetration resistance, final at 27.6 MPa, opened in the Humboldt penetrometer manuals) read through the deep punch factor 9 at 4 h and 6 h at 21 °C (the MASCO set-time table, opened), which gives $m = 5.13$ and $\tau_IS = 383$ kPa. The temperature factor $s(T)$ is the ratio of the MASCO table's final-set times. The surface layer has 15–30 % of the bulk's strength (estimate: floated to densified). Repair mortar, masonry mortar, sealant and wall compound have their own laws (the ledger, section 5): the mortar reaches "thumbprint hard" (5–20 kPa) 30–50 min after laying; the sealant has a bulk yield stress of 600 Pa and skins at 2 h (Tremco Dymonic 100, opened); the compound 700 Pa (3M US 11,267,758, opened).

The pad's timeline at 21 °C, read from the law: a boot holds at about 96 min, a fingertip at 118, a trowel stops shaping at about 210, initial set at 240.

3.4 Gravity: paste at rest and in slow flow

A paste with a free surface flows under its own weight where it yields and stops where it does not. The model is a thin-layer Bingham flux between neighbouring cells,

$q = -(\rho\, g\, Y^2\, (3h - Y) / (6\, \mu))\, \nabla h,\ Y = \max(0,\ h - \tau / (\rho\, g\, |\nabla h|))$,

computed conservatively on the faces where a **width-aware yield gate** opens. The gate asks whether the step between two cells could stand as a vertical cut of that height (the critical height of a vertical cut in a cohesive material, $3.83\ \tau/(\rho\, g)$) at several widths about the face, so that a cell-sized ridge on a stiff paste stands while a heap of soft paste slumps; the surface layer's drop is clipped to its own thickness. The flux is limited per face (an eighth of the step, a quarter of the donor's depth per sub-step) and by an inertial cap $2\sqrt{(g\ t)}$, sub-stepped to stability up to 40 sub-steps per call, and confined to the tiles near open faces. A region that nothing has touched since gravity last found it at rest is not evaluated again (yield stresses only grow, so a surface at rest stays at rest until something changes it).

3.5 Bleeding, evaporation, sealing and blisters

Concrete bleeds: water rises at a rate shaped as a triangle over the bleeding period (mean 0.15 kg/m²/h, ending when the bulk reaches 10 kPa, about 110 min at 21 °C; estimates sized to CIP 14's "sheen gone when a boot holds") and evaporates at the rate of the ACI 308 / FHWA-RD-02-099 equation for the concrete at the site's temperature and the room's air (still, door open, fan). Where the surface is sealed (S, from steel troweling while the concrete still bleeds) the rising water is trapped (q) instead of reaching the surface; trapped water above 0.2 mm raises a blister of a few centimetres (CIP 13's description, a phenomenological rule). Finishing over a standing film works it into the top 6 mm and raises its water-cement ratio (CIP 1's dusting mechanism).

3.6 Tools on the paste: the support plate

A rigid tool resting on a paste is carried by a **support plate**: one frictionless static box per tool sole (and per fingertip pad), moved every tick to lie under the sole at the level where the paste's bearing capacity carries the force the plate carried on the previous tick. MuJoCo then resolves the contact between the sole and its plate like any other, so the tool is supported, slid, pitched and lifted by the rigid solver, and the paste model decides only where the support is. The plate meets only its own sole's geometry (a free collision bit of its own) and nothing else of the world.

The bearing capacity of the cells under the sole is

$$p = \min(\ N_c\ (1 + 0.2\ B/L)(1 + 0.2\ \min(D/B,\ 2.5))\ \tau_b\ ,\ \tau_s\ (2 + B\ /\ (4\ t_s))\ )\ \cdot\ (\text{cover})$$

the lower of a bearing failure through the bulk (Prandtl's strip factor $N_c = 5.14$ with Skempton's shape and depth factors, 9.25 for a round punch 2.5 widths deep: the pocket penetrometer's factor 9) and a squeeze-out of the weak surface layer between the sole and the bulk (Prandtl's thin layer). B and L are the bearing patch's width and length (a pitched blade bears on a strip along its trailing edge, and its B is that strip's), D the depth of the sole below the paste around it, and the cover the share of each cell under the sole. When the force exceeds the capacity the plate sinks at a yield rate $(F - C)/(C_Y\ A)$ with $C_Y = 62.5$ kN · s/m³ (62,500 N · s/m³; numerical yielding resistance, not material viscosity), never faster than the sole and never below the rigid base. The plate is placed at the sole's lowest point and turns with it; its contact is soft (solref 0.01 s, damping ratio 2), which cuts the ticks without contact by two thirds in a combined check.

Two rules decide how the plate follows a moving sole. **A sliding sole climbs the paste ahead of it only as its own underside lets it**: while the sole advances (at least 5 mm/s), its plate rises no faster than the advance times the rise of the sole's underside along the advance (plus 1 mm/s), so a flat blade cuts what stands above its level and a pitched one rides up its ramp; a sole pressed in without advancing is met by the paste where it stands. And the sole's lowest point, sampled at the lattice's cell centres, lies up to a cell's run of its tilt above or below the real edge, so the plate is held within that slack of the sample, and where the solver has the sole resting on the region's rigid base the sole is put on the base. Each rule answers a failure under a carriage doing a task's work: a plate that rises at 40 mm/s carries a darby on the highest few per cent of the struck pad, and a plate pinned 0.05 mm above a sample lets a pitched trowel fall through the pad (bench B11 sets every floor tool down at four offsets from the lattice).

3.7 What the sole does to the paste

Where paste stands above the sole's surface (the sole's shape: flat, a float's bevelled end, an edger's quarter round and lip, a jointer's rod, a sealant tool's rounded nose, a knife's blade), it is **cut**. What is cut goes:

- into the **roll** ahead of a sliding sole, carried along and set down where the sole fills a low below it, the roll's prism (a right triangle against the leading face, its height H the sole's rise plus the edge's lift, so $A = H^2/2$ of cross-section, over the face's length; for an edger the rise is that of its bearing flat, about 5 mm at its 2° pitch, not the depth of its lip) holding what it can and the rest running off the roll's two ends. The face that leads is the blade's long side when it moves across its length (a float or a darby pushed broadside: the overflow stands as lap ridges beyond the blade's ends, a bead across its width fading out over 20 mm) and its end when it moves along its length on an open surface (an edger run along a form: the roll stands across the end and the overflow is left beside the blade's leading corners, as the ridge beside an edger's track that a finisher floats out; an edger's lip, down the form's face, closes the roll's end on that side). In a joint the roll of stiff mortar is pushed along the joint ahead of the jointer. The overflow stands no higher than the roll it runs off (H above the sole's lowest point, or above the paste's surface where the sole came onto it for this stroke, if that stands higher, as for a sunk sole): what does not fit in the local 20 mm band stays in the roll. When the sole reverses or leaves the

paste, the roll stays where it stood, as the same prism ahead of the face that led (the long face when the blade moves across its length, the end when it moves along it);

- for a tool that **carries** (knives, the margin trowel, the sealant tool), onto the blade, up to the blade's load (an estimate per tool); beyond it the cut runs off as a roll's excess does, and the load stays on a lifted blade while the paste's yield stress holds it there;
- under a sole **pressed in place** (advancing slower than 5 mm/s), up around the footprint as a bearing failure heave (a band as wide as the sole's narrow side, between 6 and 60 mm, Prandtl's passive wedges, which rise only in paste: not through a form, the slab or a wall beside the sole). The advance is the sole's in-plane velocity low-passed twice over 0.1 s (a 1 mm, 8 Hz hand tremble, 50 mm/s at its peaks, passes as 1.9 mm/s; a stroke at 80 mm/s counts as advancing after 40 ms): a sole that trembles in place while it sinks presses its paste out around it and builds no roll, and a tremble's reversal is not a stroke's;
- in **mortar**, normalized surface-fabric consolidation C accumulates confined interface work and increases surface resistance; no assumed air fraction deletes paste from the mass ledger;
- in a **sealant joint**, out over the joint's edges where the tool does not carry it.

Whatever a tool lays down (a roll set down, an overflow, a heave, a squeeze-out) is spread over its whole footprint on the region's 1–2 mm lattice, past the region's edge too, and every cell gets its share of the whole footprint. The share past a grid boundary is deposited on sparse adjacent wall/slab tiles, with its location, material history and contact geometry. A tile is 128 cells across and inherits the parent grid spacing. The contamination criterion reads the **current** remaining catch volume. A knife can collect it; no counter is cleared by cleanup. Falling packets land on horizontal work surfaces or the floor. These supports are bounded approximations: packets do not collide with every pipe, cart and robot link, and flux across catch-tile edges is not coupled.

A sliding sole also **works** the surface under it (only above 1 mm/s, so that a sole trembling in a hand does not): it embeds the coarse aggregate showing, densifies and seals the surface by its steel's factor (steel trowel 1, magnesium float 0.3, wood float 0.2), leaves its texture (a float's swirl roughness, a trowel's smoothness), and works a standing water film in. Embedding and texture act per pass of the bearing patch over a cell: with n = ds/B the passes in a tick (ds the distance slid, B the bearing patch's extent along the motion, a pitched blade's strip of a few millimetres or a flat sole's width),

$$E \leftarrow E \cdot \exp(-n \cdot k_E \cdot \mathrm{clip}(p/1\ \mathrm{kPa}, 0.1, 20)),\quad a \leftarrow a_\mathrm{tool} + (a - a_\mathrm{tool}) \cdot \exp(-n \cdot k_a \cdot y),$$

with p the bearing pressure, y the share of the surface layer's yield the pressure reaches, $k_a = 2.2$ a pass and $k_E = 8.3$ a pass at 1 kPa (two passes of the darby lying flat on its own weight, 0.164 kPa, take the struck surface's 30 % of exposed aggregate below 2 %: CIP 14's "the bull float or darby embeds large aggregate", the count of passes an estimate). Densification is per metre slid, as the work the interface drag does. Integrated per metre slid, embedding and texture would do a thirtieth of a pass under a pitched blade, and the darby would need some fifty passes.

The paste pushes back: an interface drag against the sole's sliding (a share of the surface yield stress over the bearing area, from a wet film to a stiff surface, bounded by the bearing force, Green's combined-load relation), a ploughing drag where the sole's edge cuts, the roll's drag, and a **tack** when a sole that has left its support is lifted off wet paste (0.7–3.9 kPa over the last bearing area, with an estimated linear falloff over 3 mm; the cited mortar test supplies only a force-scale anchor, not this separation law). All are applied to the tool's body at the sole, regularised for explicit stability.

3.8 Finite surface composition and material transport

Initial weak paste plus an immediately underlying convertible reserve occupy at most `T_SURF_MAX` (nominally 3 mm). Working transfers thickness from the finite reserve into the weak surface layer. Cutting removes surface paste and reserve in depth order. Newly exposed bulk receives no new reserve. This bounds repeated weakening without imposing a maximum trench depth. The layer budget and 0.15–0.30 strength ratio remain estimates, checked over 1–3 mm and ratios spanning 0.15–0.60 in the coupled sensitivity cases.

A carried `Packet` stores its actual material configuration, volume, age, surface fraction, reserve, water, densification, texture, wet-work and timing history. Receiving cells mix intensive history by volume: `q_new = (V_old q_old + V_in q_in)/(V_old + V_in)`. Empty cells take source history. Gravity advects these histories with its thickness flux. This is a mixed-column approximation, not stratified chemistry; different material species are not silently transmuted.

Every carried pose receives weight `rho V g`, including travel outside all material regions. Retention compares mound stress to the carried material's current yield stress; orientation reduces capacity toward a 2% inverted adhesion allowance. That allowance is estimated, not a measured bond law. Detached packets follow ballistic motion to the modeled supports and leave visible, collectible deposits. Load meshes are volume-scaled ellipsoids, not resolved free-surface flow.

Blister elevation is a geometric envelope, excluded from solid-paste volume. Cutting collapses its void before transferring paste. Mortar consolidation updates history without destroying an untracked paste volume. Total paste is counted over roots, catch tiles, tool loads, falling packets and the rag.

3.9 Skin, open mesh and flexible knife

Sealant work is compared with **material age at contact**, including initial age and transferred history. At skin onset the surface resistance is multiplied by three; sliding tears the skin, retains roughness and cannot create new joint-wall contact. Bulk yield remains 600 Pa. This is a minimal estimated skin model, not calibrated polymer fracture. The fast variant's 20-minute deadline is tested at 19 and 25 material minutes. Applying the concrete temperature table to sealant is explicitly an extrapolation.

The wallboard mesh has a 0.5 mm contact envelope but only 30% solid fibre volume. Compound occupies the remaining openings. Separating those quantities matters: using the volume-equivalent 0.15 mm base as contact height let the knife scrape the mesh bare. The visible weave is an approximate geometric representation of the same envelope/solid fraction.

The knife's contact surface follows an Euler–Bernoulli cantilever shape, `delta(xi) = (F/K) xi²(3-xi)/2`, clamped at the handle. `K = 2800 N/m` at 152 mm blade width scales with width; deflection is limited to 3 mm. The body/grip stays rigid. The support plate compensates for this displacement and uses the continuous rigid blade edge instead of a phase-dependent grid sample. Individual fibres, shell dynamics and distributed edge flexure are not resolved. A separately labeled finite-force grade fixture demonstrates coverage/feathering; this assistance is not available as a policy action.

3.10 Wait, interaction scope and numerical choices

Rigid dynamics use 0.5 ms ticks. A legal accelerated material wait advances only material age/flow, then performs 0.2 s of full reseating. It is refused while bodies move, targets remain active, a tool is loaded against a region, or a material packet is falling. Performance denominators below use only the actual rigid observation clock.

Only nominated soles and fingertip pads receive active paste contact. The full robot, every dropped-tool side and edge-on/back-facing soles are not qualified against wet fields. The H-4134 is a rigid force probe with robot reaction observations, not a spring instrument with a retained reading ring. Per-tick support travel, soft-contact damping, velocity filtering and flux caps are numerical choices, not measured material constants.

3.11 Mortar surface consolidation and bounded rate assumptions

Mortar tooling is modelled as a normalized surface-fabric consolidation C in [0,1], **not** a porosity or lost-volume fraction. With actual normal pressure p, slip ds, interface factor m, bulk yield tau, layer depth delta=2 mm and characteristic strain gamma=6:

```
dW = m tau ds                          [J/m2]
W* = tau delta gamma                   [J/m2]
C' = 1 - (1-C) exp[-p/(p+tau) dW/W*]
tau_surface = tau (1 + 0.30 C)
extra bearing = N_c (tau_surface-tau) min(delta/B,1)
```

Drag reads the changed surface yield; bulk strength remains age controlled. Paste density/volume are unchanged and C travels with material packets. C=0.5 means half of the estimated mobilizable surface response, giving 15% nominal surface hardening. It does not predict permeability or cured strength. BIA TN 7B supports consolidation by tooling at thumbprint hardness, not these coefficients. Layer depth 1–3 mm, characteristic strain 3–12 and hardening 0.15–0.45 are explicit uncertainty ranges. At the 37-minute scenario strength, W* is about 125 J/m2.

Native prescribed-work sensitivity checks vary texture rate 1.1–3.3, embedding 4.15–12.45, densification energy 56–224 J/m2, interface shear and surface strength. The densification energy of 112 J/m2 is named in the ledger; it is equivalent to 56 kPa of plastic work over 2 mm, not a measured viscosity. Numerical plate damping and travel filters are separate contact stabilizers.

4 Shared robots, controls, observations and evaluation

The default is `mobile-fr3-duo`; `mobile-fr3-single` builds one arm. Both come from the shared catalog and versioned robot models/rigs. Duo/single live inventories are 28/19 joints, with two/one arms. Equivalent validated robot JSON is supported; incompatible models, mounts, attachments and missing-arm commands fail explicitly. There is no humanoid configuration. Low work height, tool loads and the absence of qualified kneeling/balance controls are the reasons; no full three-family qualification is claimed.

Policy actions are base twist, named joint targets, Cartesian arm deltas, gripper width/force commands, keyboard chords and the constrained raw material wait. Physics/evaluation/checkpoints and diagnostic views are owner-only. Observations provide robot state, policy cameras, task instructions and action limits; constitutive state is not leaked to the policy. The HTTP/dashboard contract, reset, shutdown and policy/owner separation remain the shared GAR interfaces.

The nine task names are `pad-darby`, `pad-edge`, `pad-float`, `pad-trowel`, `pad-finish`, `patch-flush`, `tool-joints`, `tool-sealant`, `patch-drywall`. Evaluators read surface height/straightedge error, texture/densification, edge profiles, water damage, joint shape/compaction/timing, wall contact, holes, mesh coverage, contamination and robot housekeeping. A local component recording is not a complete task score. The mortar criterion measures normalized fabric C >= 0.5.

5 Grounding and parameter choices

The ledger below distinguishes opened sources, derived values, scenario choices and estimates. The concrete initial yield follows NISTIR 6094 equation 19. The conversion of penetration resistance through a factor of nine into the whole age/yield curve remains an assumption; initial/final-set anchors do not uniquely determine it. The repair mortar is a generic scenario blend of density and work/set-time anchors from different products, not a calibrated named product.

NRMCA CIP 14 supports sequencing, aggregate embedding and avoiding bleed water; it does not supply the per-pass rates, 112 J/m² densification energy, interface-shear factors, mortar compaction/window or acceptance thresholds. `EMBED_PASS` partly fits an assumed two-pass outcome. The sensitivity table checks that the repaired cutting behavior survives the stated combinations; it is neither a full uncertainty study nor validation of all these coefficients.

The tack source used a 40 mm diameter specimen of cellulose-modified cement–lime–sand mortar, compressed initially to 3 mm, rested three minutes and pulled at 10–500 micrometres/s. The **initial thickness is not a detachment distance**. Force estimates 0.85–4.9 N divide by `pi (0.020 m)²` to give about 0.676–3.899 kPa. The model's rounded force scale, 3 mm linear falloff, speed interpolation/extrapolation and cross-material multipliers are separate estimates. A bench reproducing its inserted law is implementation evidence, not independent calibration.

Name	Value	Unit	Label	Basis
ROOM	(6.1, 4.88, 3.05)	m	estimate	a small pump room, 20 x 16 ft, deck at 10 ft
WALL_T	0.194	m	opened	8 in CMU, 7-5/8 in actual (standard nominal block; BIA TN 10 3/8 in joints)
PARTITION_T	0.124	m	estimate	3-5/8 in steel studs with one layer of 5/8 in board each side
DOOR	(4.6, 5.51, 2.13)	m	estimate	hollow metal door 36 x 84 in in the south partition (x from, to, height)
MU_FLOOR	0.5	-	estimate	steel-troweled sealed slab against painted steel and wood
PAD_CENTRE	(3.2, 2.9)	m	authored	room layout
PAD_SIZE	(1.219, 0.914)	m	authored	48 x 36 in inside the forms
PAD_T	0.089	m	estimate	a '4 in' pad formed with 2x4s on edge is their dressed 3-1/2 in
FORM	(0.038, 0.089)	m	opened	2x4 dressed 1-1/2 x 3-1/2 in (standard lumber size)
KICKER	(0.15, 0.038, 0.089, 0.6)	m	estimate	2x4 blocks fastened to the floor every 0.6 m outside the forms
FORM_DISTURB_N	150	N	estimate	lateral load on a braced form that would move it
PATCH_CENTRE	(4.6, 1.25)	m	authored	where the old pad was broken out
PATCH_SIZE	(0.4, 0.25)	m	authored	room layout
PATCH_DEPTH	(0.025, 0.038)	m	opened	SikaQuick Pro 1000: at least 1/4 in, neat to 2 in; chipped 25-38 mm (estimate)
PATCH_MARGIN	0.2	m	derived	slab modelled around the cut: the 406 mm wood float laid across the 250 mm patch overhangs the cut by 78 mm a side and spills 20 mm beyond its ends; a 50 mm margin puts both spill ends off the modelled slab
PATCH_SAWCUT	0.013	m	estimate	square saw-cut edge 1/2 in deep (ICRI practice; not opened)
BLOCK	(0.397, 0.194, 0.194)	m	opened	8 x 8 x 16 in CMU: 15-5/8 x 7-5/8 x 7-5/8 in actual
JOINT_W	0.0095	m	opened	3/8 in mortar joint (BIA TN 10)
INFILL_X	(3.89, 5.11)	m	authored	old duct opening, three blocks a course
INFILL_ZO	0.813	m	authored	sill at four courses (4 x 203 mm)
INFILL_COURSES	4	-	authored	813 mm opening
JOINT_BED_DEPTH	0.02	m	estimate	mortar joint depth modelled behind the block face (face shells 32 mm)
CJ_Y	2	m	authored	control joint in the east wall
CJ_W	0.013	m	opened	1/2 in sealant joint (Sikaflex-1A: 2:1 width to depth)

Name	Value	Unit	Label	Basis
CJ_BACKER	0.0085	m	derived	backer rod face 8.5 mm behind the wall face: 6.5 mm of sealant at the tooled centre + 2 mm of concavity
CJ_SEGMENT	(0.45, 2)	m	authored	task segment between two strips of painter's tape
DW_PATCH	(1.8, 1.25, 0.038)	m	authored	centre x, z of the mesh-patched 1-1/2 in hole
DW_HOLES	((1.52, 1.42), (1.6, 1.42), (2.02, 1.08), (2.1, 1.08))	m	authored	anchor holes of two removed conduit straps (x, z)
DW_HOLE_D	0.006	m	estimate	1/4 in plastic anchor holes
BOARD_T	0.0159	m	opened	5/8 in gypsum board (standard)
MESH_T	0.0005	m	estimate	self-adhesive fibreglass mesh patch
MESH_SOLID	0.3	-	estimate	homogenized fibre area/volume fraction; USG J780A requires compound forced through open mesh; not a measured weave
RHO_ CONCRETE	2350	kg/m3	estimate	normal-weight concrete
SLUMP	0.1	m	opened	CIP 13: 3-5 in (75-125 mm) to avoid blisters
TAU_BO	1566	Pa	derived	NISTIR 6094 eq. 19 at slump 100 mm (opened)
A_THIX	0.25	Pa/s	estimate	inside Roussel's ordinary range: < 0.1 non-thixotropic, > 0.5 highly (opened)
N_DEEP	9	-	estimate	deep circular punch bearing factor (Skempton; snippet/memory); a factor of 6 would move set times 8 %
PR_IS	3.45e+06	Pa	opened	ASTM C403 initial set, 500 psi (Humboldt H-4134 manual)
PR_FS	2.76e+07	Pa	opened	ASTM C403 final set, 4000 psi (Humboldt H-4137 manual)
T_IS	1.44e+04	s	estimate	initial set 4 h at 70 F (Humboldt: first test at 3-4 h)
T_FS	2.16e+04	s	opened	final set about 6 h at 70 F (MASCO table)
SET_M	5.129	-	derived	power of time between the two set anchors (5.13)
TAU_IS	3.83e+05	Pa	derived	yield stress at initial set (383 kPa)
MASCO	30 19; 40 14.67; 50 10.33; 60 8; 70 6; 80 4; 90 2.667; 100 1.667	F: h	opened	approximate final set against temperature (MASCO forming catalogue 2016)
R_SURF	(0.15, 0.3)	-	estimate	surface layer / bulk strength, floated -> densified
T_SURF_MAX	0.003	m	estimate	paste layer after floating (CIP 20 delamination 3-6 mm, CIP 1 top 1/4 in)
T_SURF_ STRUCK	0.0008	m	estimate	paste layer of a struck-off surface
MU_BULK	50	Pa s	estimate	plastic viscosity of slump concrete; only slow flow reads it
MU_SURF	5	Pa s	estimate	plastic viscosity of the paste layer
CEMENT	350	kg/m3	estimate	cement content, for the water worked into the top 6 mm
WC	0.5	-	estimate	water-cement ratio of the pad's mix
DWC_DUST	0.05	-	estimate	rise of the top 6 mm's w/c counted as a dusting risk (CIP 1 mechanism)
AGG_EXPOSED	0.3	-	estimate	share of a struck-off surface with coarse aggregate showing
EMBED_PASS	8.3	1/pass at 1 kPa	estimate	e-folds of aggregate exposure per pass of a bearing sole at 1 kPa (linear in pressure, 0.1-20 kPa): two passes of the 1.3 kg darby lying flat on its own weight (0.164 kPa) take the struck surface's 30 % below 2 % (CIP 14: the darby 'embeds large aggregate'; two passes an estimate). As a sliding distance (0.25 m at 1 kPa) it would need some 50 such passes
TEX_PASS	2.2	1/pass	estimate	e-folds of the surface texture toward the tool's own per pass of a bearing sole where the surface yields fully (as a 40 mm sliding length gives an 89 mm float bearing over its whole width)

Name	Value	Unit	Label	Basis
DENS_ENERGY	112	J/m2	estimate	e-fold surface densification work; equivalent to 56 kPa plastic work density over 2 mm; test 56-224 J/m2, not a CIP/ASTM measurement
BLEED_MEAN	0.15	kg/m2/h	estimate	0.15 gives CIP 14's order (sheen gone after the boot holds); FHWA paving eq. 0.09, general use 0.5-1.5 (opened)
BLEED_END_TAU	1e+04	Pa	estimate	bleeding ends when the bulk reaches 10 kPa (about 110 min at 21 C)
SHEEN_W	2e-05	m	estimate	a film thicker than 0.02 mm reads as sheen (render and evaluator)
AIR	still (20, 50, 0.2); door (20, 50, 1); fan (20, 40, 2.5)	C, %, m/s	estimate	room air at the slab: still, door open, a pedestal fan left running
BLISTER_Q	0.0002	m	estimate	trapped bleed water that raises a blister (CIP 13 sizes 12-100 mm, skin 3 mm)
REPAIR	rho 2051; tau0 2000; t_is 3000; mu 20; bleed 0.05	kg/m3, Pa, s, Pa s, kg/m2/h	scenario blend / estimate	generic repair mortar: Quikrete density 2051 kg/m3 combined with SikaQuick 35-85 min set and 30 min finishing; not a product-specific material
MORTAR	rho 2000; tau0 1000; tau_ref 8000; t_ref 2100; m 3; mu 10	kg/m3, Pa, Pa, s, -, Pa s	estimate	Type N mortar stiffening by suction: thumbprint hard 5-20 kPa about 30-50 min after laying
THUMBPRINT	(5000, 2e+04)	Pa	estimate	a thumb (20 N on 2 cm2) prints without pick-up (BIA TN 7B rule)
MORTAR_COMPACT	1	-	model definition	normalized mobilized surface fabric, NOT a lost volume or air fraction; compact=1 is saturated surface consolidation, task requires >=0.5
MORTAR_LAYER	0.002	m	estimate	surface layer rearranged by tooling; order of mortar sand grains; sensitivity 1-3 mm, not whole 20 mm joint depth
MORTAR_STRAIN	6	-	estimate	characteristic accumulated shear strain for surface fabric rearrangement; W=*tau_bulk*layer*strain; sensitivity 3-12
MORTAR_HARDEN	0.3	-	estimate	maximum surface yield increase from mobilized fabric; bounded sensitivity 0.15-0.45; not a compressive-strength or permeability prediction
SEALANT	rho 1400; tau0 600; mu 50; skin 7200	kg/m3, Pa, Pa s, s	estimate / opened	non-sag bound 180 Pa for 13 mm (derived); skin 2 h (Tremco Dymonic 100)
COMPOUND	rho 1000; tau0 700; mu 6	kg/m3, Pa, Pa s	opened / estimate	3M US 11,267,758: 693-753 Pa (Ex. 31, 20), 5.8 Pa s at 1.5 kPa; density estimate
NC	5.142	-	opened	Prandtl's strip factor; Green's combined-load factor (Femmam et al. 2018 Table 1)
TACK	(700, 3900)	Pa	derived / estimate	graph force anchors 0.85-4.9 N divided by pi*(0.020 m)^2, rounded; cellulose-modified cement-lime mortar, initial 3 mm thickness, 10-500 micrometre/s separation (Le Thanh Hai & Phan Van Tien 2014); cross-material extrapolation estimated
TACK_GAP	0.003	m	estimate	linear detachment distance, not measured: Le Thanh Hai & Phan Van Tien's 3 mm is INITIAL specimen thickness, not detachment length; cross-material multipliers and robot lift-rate extrapolation are estimates
M_INTERFACE	(0.05, 0.3)	-	estimate	interface shear / surface yield stress, wet film -> stiff (open)
PLOUGH	3	-	estimate	work per volume ploughed by a sole's edge, in units of the yield stress
K_CONTACT	0.15	-	authored	paste contact stiffness = this x tool mass / dt^2 (explicit stability)
C_CONTACT	0.3	-	authored	paste contact damping = this x tool mass / dt
ROLL_LIFT	float 0.003; trowel 0.001; knife 0.001	m	estimate	how far a bevelled or turned edge lifts the roll
DROP_STICK	6	-	estimate	a load stays on a lifted blade while rho g h < tau / this

Name	Value	Unit	Label	Basis
TOOLS	(see section 2.3)	m, kg	opened / estimate	sizes from the maker's pages; masses = shipping weight less 10-25 % (estimate); the penetrometer from the Humboldt H-4134 manual (opened: needle 1/20 sq in, body 3/4 in x 7 in, 8 oz); the sealant tool's 5/8 in width is the Albion classic spatula size next above the 1/2 in joint (opened: chemical-concepts.com, sizes 3/8 to 1-1/2 in), so it rides both joint edges; its rounded nose, 11 mm in radius across the blade, is an estimate that stands 2.1 mm into a 13 mm joint between the edges ('tooled flush to slightly concave', NIH ORF News to Use, March 2021, opened)
CARRY_CAP	sealant_tool 1.5e-05; putty_knife 2.5e-05; joint_knife 0.0001; margin_trowel 0.00015	m3	estimate	paste a loaded blade holds (the sealant tool: a gob about 30 mm across riding ahead of its nose; a knife's or margin trowel's face); what a sole cuts beyond it runs off over the joint's edges or the blade's ends. The tool-sealant bead displaces about 45 ml per metre when tooled (B7), so a finisher wipes the tool every 0.3 m
RAG	(0.25, 0.25, 0.012)	m	estimate	a folded shop towel on the cart top (a rag goes with the sealant tool)
RAG_WIPE	0.05	m	estimate	rubbing a loaded blade on the rag over this distance leaves 1/e of its load on it
EDGER_TRACK	0.0012	m	estimate	the edger's flat pressed this far below the field leaves a step at its track's inner edge (floated out after)
FLEX_K	2800	N/m	estimate	6 in flex joint knife at its edge: 0.5 mm spring steel, 70 mm free
SKIN_FACTOR	3	-	estimate	minimal post-skin surface resistance multiplier, not polymer calibration; tooling tears the skin and cannot establish new wall wetting
CART	(0.86, 0.46, 0.84, 15)	m, kg	estimate	three-shelf utility cart (length, width, top height, mass)
TUB	(0.9, 0.6, 0.2)	m	estimate	mixing tub with leftover concrete
HAWK	0.33	m	opened	Marshalltown 1D aluminium hawk 13 x 13 in
MORTAR_BOARD	(0.749, 0.737)	m	opened	Marshalltown mortar board 29 x 29-1/2 in
EVAL	(see section 4)	-	authored	the evaluator's thresholds
SLOW_DT	0.05	s	authored	material clock step while the robot works (strength, bleeding, slow flow)
WAIT_DT	1	s	authored	material clock step of a wait command
WAIT_MAX	3600	s	authored	longest wait command
REST_SPEED	0.001	m/s	authored	a wait needs every free body slower than this

6 Component experiments

All component films use finite phantom forces/torques and privileged initial placement unless labeled unassisted. Scoop/carry begins with an unloaded knife and uses actual material pickup. Rag and spill fixtures explicitly start with authored 1 ml and 5 ml loads. Troweling begins on a locally level, already floated patch with 0.15 mm residual ridges. Grade assistance and force caps are disclosed by the fixture. Frames are exact recorded states at 10 Hz, with a final 0.1 s packaging hold; no interpolated physics or successful-score splice is used.

Component	Rigid seconds	Fault	Exact-state film	Scope
float	3.50	none	yes	assisted; see record
trowel	6.60	none	yes	assisted; see record
patch	5.10	none	yes	assisted; see record
joint	6.50	none	yes	assisted; see record
sealant-before	2.90	none	yes	assisted; see record
sealant-after	2.90	none	yes	assisted; see record
drywall	3.70	none	yes	assisted; see record
carry	8.90	none	yes	assisted; see record
overturn	3.80	none	yes	assisted; see record
spill	2.80	none	yes	assisted; see record
rag	2.50	none	yes	assisted; see record
rest	10.00	none	yes	unassisted

The results of sections 6.1 to 6.3 were recorded on an earlier version of the paste model and renderer; sections 6.4 and 6.5 have the later components and the finish optics.

6.1 Coupled cutting and local finishing

Pitch (deg)	Press (N)	Speed (m/s)	Layer (mm)	Strength ratio	Grid shift (cells)	Trench (mm)	Floor slope
-3	30	0.15	3.0	[0.15, 0.3]	0.0	1.18	-0.0002
0	30	0.15	3.0	[0.15, 0.3]	0.5	0.59	0.0004
3	30	0.15	3.0	[0.15, 0.3]	0.0	1.02	0.001
5	30	0.15	3.0	[0.15, 0.3]	0.0	2.31	0.0006
10	30	0.15	3.0	[0.15, 0.3]	0.0	3.58	0.0006
5	15	0.08	1.0	[0.15, 0.3]	0.5	1.05	0.0005
5	60	0.2	3.0	[0.3, 0.6]	0.5	1.0	0.0003
0	10	0.2	1.0	[0.3, 0.6]	0.0	0.51	0.0005

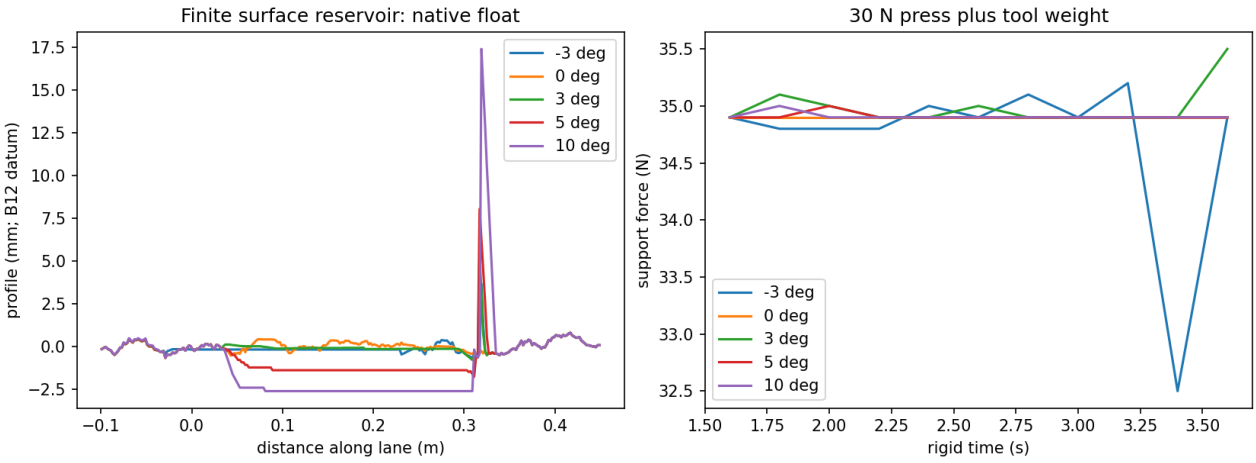


Figure 4. Native float profiles and support force from the same eight-case sensitivity records. Positive pitch is leading-edge down. The model digs under poor technique, but the finite reserve stops a sustained trench descent.

Without the finite surface reserve (section 3.8), a float pitched 5 and 10 degrees digs 12.52 and 33.79 mm (41.27 mm at the deepest point); with it, 2.31 and 3.58 mm, with floor slopes 0.0006. Their volume errors are numerical roundoff. These are model results, not measured real-concrete depth tolerances. Layer thickness, strength ratio, pressure, speed and grid phase are jointly varied; the eight runs are not a factorial identification experiment.

Interior ROI	Height range (mm)	Median texture (mm)	Median densification	Smooth-hard share
float	2.397 → 0.214	0.400 → 0.375	0.000 → 0.072	0.0%
trowel	0.296 → 0.000	0.250 → 0.103	0.200 → 0.999	0.0%

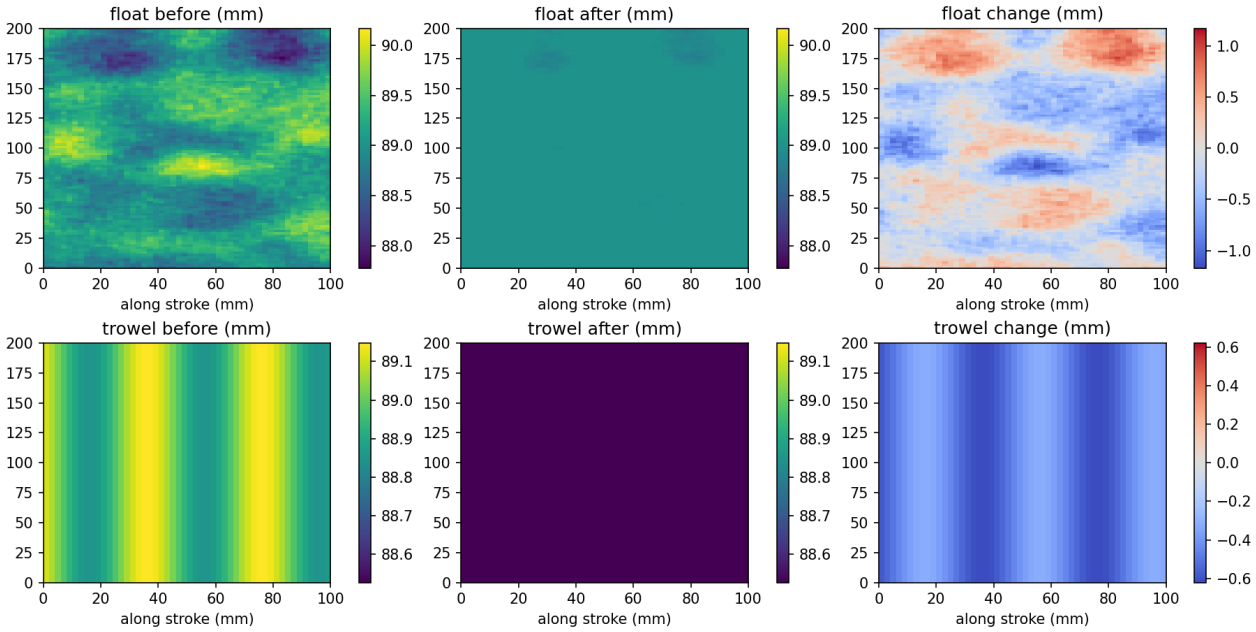


Figure 5. Before, after and difference on predefined central 100 by 200 mm areas of the float and trowel recordings. Shared before/after colour scales and separately labeled differences make the millimetre changes inspectable.

This two-pass trowel densifies and smooths the local field, but misses 0.05 mm; a six-pass sample in section 6.5 reaches it. The floor patch is struck flush and receives float texture; its six local criteria pass while visible excess remains on the slab. The large remaining roll/pile and joint tags are retained limitations, not a completed cleanup claim.

6.2 Transport, cleanup, mesh and skin

Check	Measured result
Actual scoop/carry	peak 21.236 ml; final 10.920 ml on tool
Demonstrated first anchor hole	minimum surface -15.900 → 0.000 mm; final max 0.000 mm
Overturned control	final load 0.425 ml; dropped material remains on supports
5 ml spatial spill	4.084 ml collected; 0.916 ml remains
1 ml rag wipe	0.0465 ml remains on tool
Mesh	bare share 0.0000; min/median/max surface 0.500, 0.500, 0.623 mm

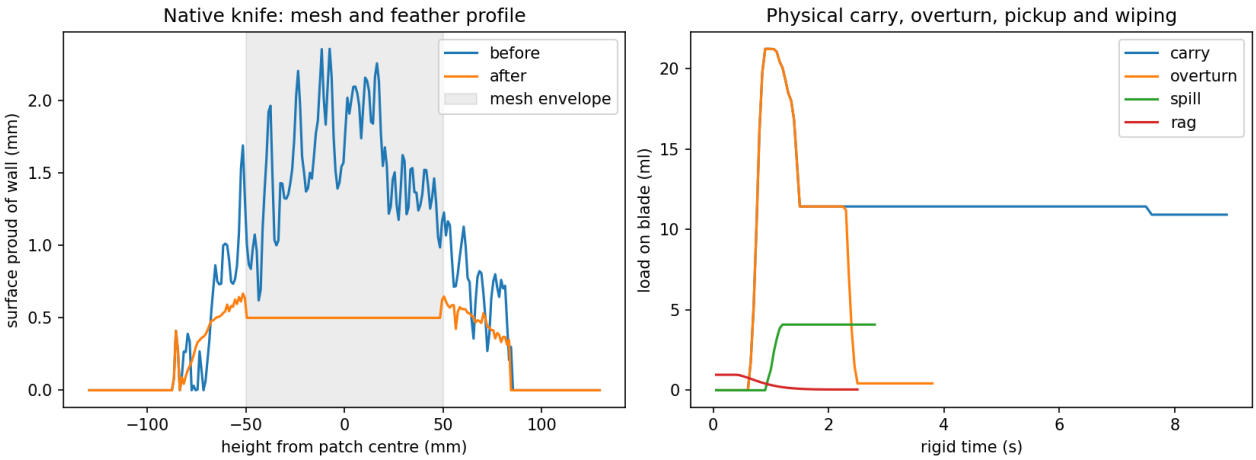


Figure 6. Native knife boundary profile and carried-load traces from the same saved states. Material deposited in the hole comes from the hawk. The separate spill/rag examples disclose their authored starting load.

The mesh is covered and the graded pass leaves a transition beyond its edge, but the entire authored wall patch is not finished: the other holes and side margins are untouched. Local graded coverage is not proof of a successful freehand force-only finishing policy.

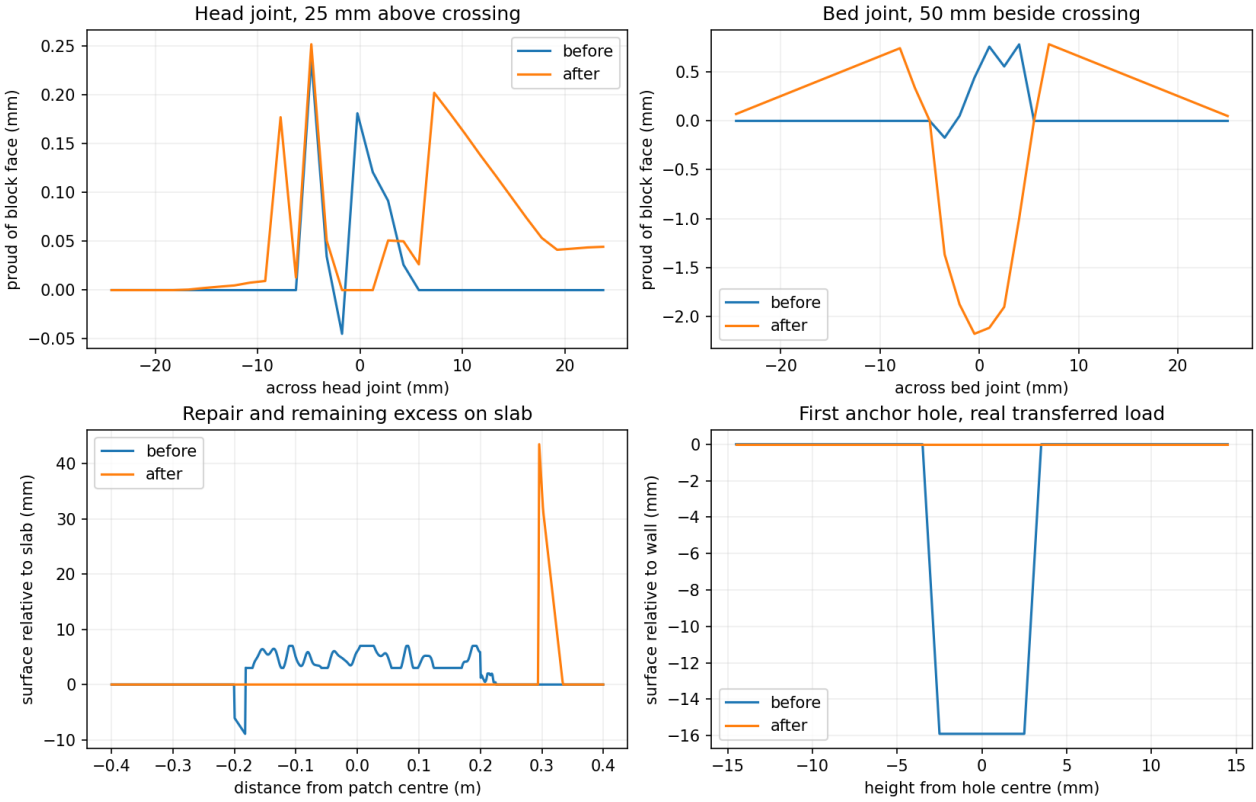


Figure 7. Fixed sections through the worked joints, floor repair and the first anchor hole. The hole reaches the wall plane using compound physically scooped from the hawk. Joint tags and floor excess remain visible in the data as well as the movies.

A close-up film of the carry uses its own diagnostic camera, with the partition's anchor holes drawn at their native 6 mm diameter and 15.9 mm depth. The joint fixture traverses 45 mm above to 45 mm below the bed joint.

Stroke	Material age (min)	Torn worked cells	Mean worked texture (mm)	Before-skin criterion
sealant-before	19.048	0.0%	0.194	True

Stroke	Material age (min)	Torn worked cells	Mean worked texture (mm)	Before-skin criterion
sealant-after	25.048	95.8%	0.495	False

Skin changes both the evaluator and physical response: the post-skin stroke tears the surface and does not newly wet joint walls. Both runs deliberately tool only a short segment and leave excess, so neither claims a complete sealed joint.

6.3 Rest and legal actions

Measurement	Result
Rigid duration	10.0 s after 1.0 s reset settle
Peak translation	0.01979 mm
Peak rotation	0.00184 deg
Peak joint coordinate drift	2.45e-05 (SI coordinates)
Fault/warnings	None / []

The rest video begins after the ordinary reset settle and contains 10 actual rigid seconds with no fixture or robot commands. Translation, quaternion-angle and joint-coordinate drift are sampled every 50 ms; every recorded pose and mesh is finite. This does not exclude high-frequency motion between samples or prove every seed over hours. A rest check on an earlier version of the code (B1) covers all nine seed-0 starts for about 12.2 rigid seconds plus material-only waiting.

6.4 Pad and seam components and finish optics

Component	Rigid seconds	Peak assist N	Volume error ml	Fault
joint	15.2	35.07	-1.7e-12	None
darby	3.5	82.87	0	None
edge	4.0	32.27	0	None
seam	2.8	13.79	0	None
seam-shift	2.8	13.79	0	None

The default renderer reads the exported texture attribute. Deterministic world-space grain gradients at estimated 3 mm and 1.3 mm scales perturb the shading normal; pixel derivatives filter unresolved detail and amplitude broadens the highlight. Geometry, paste volume and color are unchanged. Three broad ceiling/work-light directions expose texture on horizontal and vertical stations. These optical choices communicate the modeled field; they are not a calibrated roughness instrument or a fracture mesh.

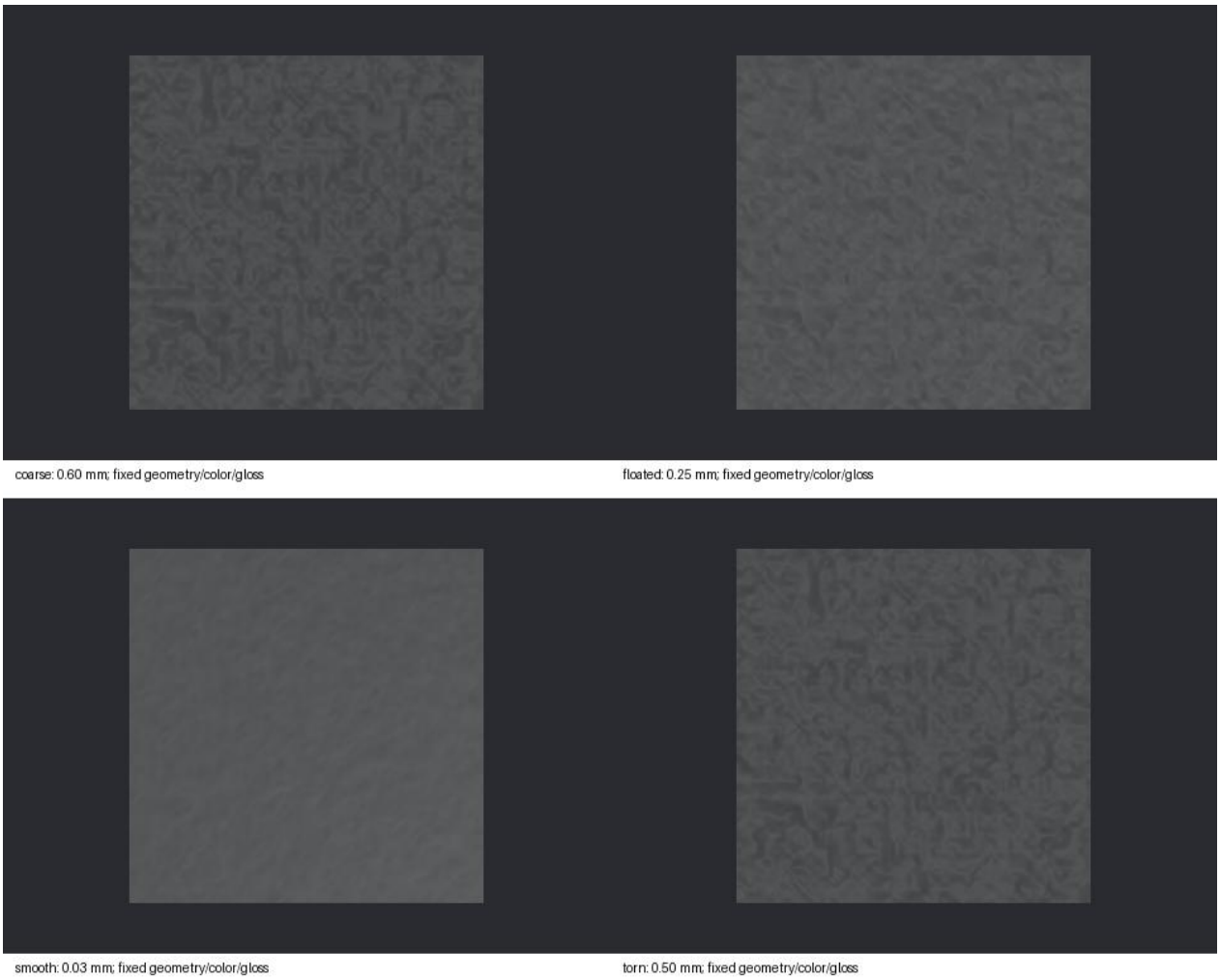


Figure 8. Native default renderer, fixed pose/geometry/color/gloss, four physical texture amplitudes. These authored counterfactuals isolate observation of the field; they are not four simulated finishing trials.

The interpolated noise is normalized for its amplitude attenuation: the 0.7/0.3 mixture has approximate RMS 0.1408 before normalization. The mapped amplitude describes RMS microrelief, and sealant keeps its half-amplitude mapping. The fixed-state images visibly separate floated and smooth finishes: their demeaned spatial difference is 3.23 channel levels, against 1.03 without the normalization; that statistic is supporting evidence, not a visibility acceptance rule. Geometry and material state are unchanged.



Figure 9. The before- and after-skin sealant states of section 6.2 after tool withdrawal, rendered through the normalized optics. Rough relief is visible in the late-worked strip.

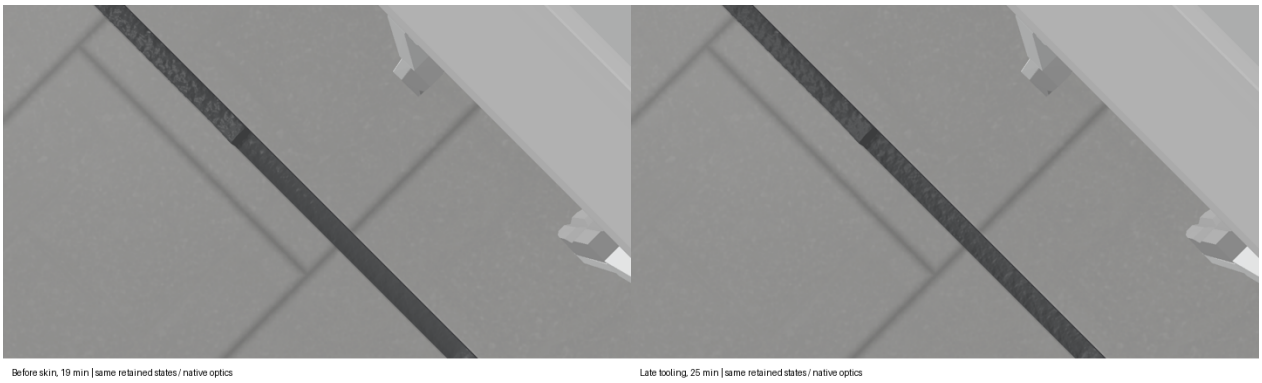


Figure 10. Actual left-wrist camera at a robot-reached pose, 207 mm from the wall, normal 87-degree intrinsics at 1280x720. Recorded arrays are inserted at the same pose for optical comparison. Fingers partly enter the frame but do not cover the inspected strip.

The wrist positioning uses privileged path planning and normal joint commands. This shows feasible optical access, not a policy finishing or full path-stability qualification.

A local contact window combines root and adjacent catch cells with unique ownership. One support calculation and one cut/fill/work/drag operation see the entire sole footprint; all material histories scatter back to their owners. New material allocates real catch cells. The interior fast path remains. Gravity flux across separate catch tiles is outside the model.

Six prescribed-pose cases vary deposit/grid phase and allocation order; two more check floor catches and snapshot restoration. The blade collects all 1.238 ml above its loaded compliant sole, with identical aggregate support area/capacity and 1800 s source age. This check includes plate placement and native knife deflection. Two actual native-knife recordings scrape an authored 5 ml wall deposit spanning four catch tiles, once shifted 37 mm. Each collects 4.999999 ml, with volume error at roundoff. Their films show the deposit disappearing into a visible carried load; no contamination counter is cleared.

Mesh coverage means reaching the 0.5 mm fibre-top envelope within 0.01 mm, not merely containing 0.05 mm of compound in the openings. A synthetic 0.21 mm total-height control fails. The graded knife record has zero underfilled mesh cells under this rule (minimum height 0.500 mm).

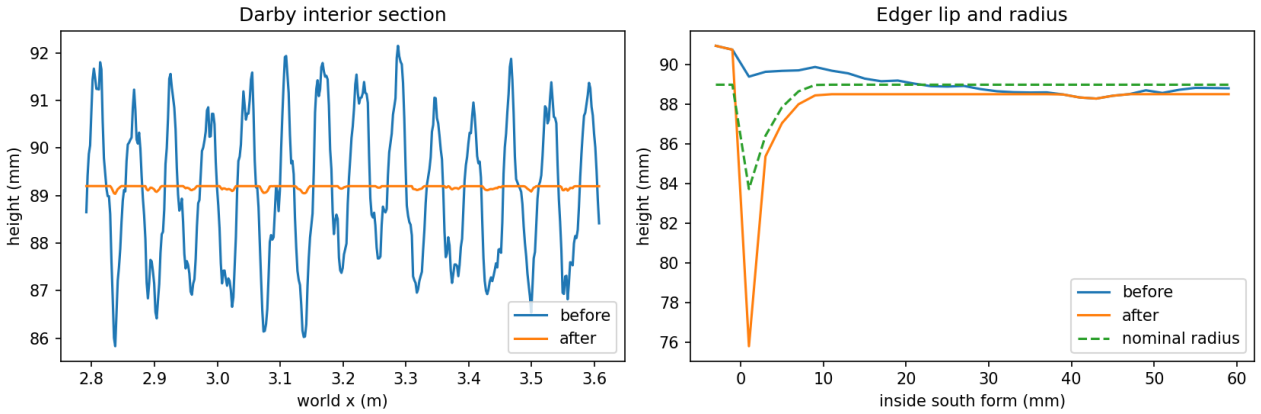


Figure 11. Native darby and edger sections. The darby's finite position carriage follows form-top grade. The edge fixture traverses one south-form segment with a modest 1-degree leading-end dip.

The darby's predefined central ROI height range falls from 10.28 to 1.83 mm, while mean exposed-aggregate state falls from 0.300 to 0.207. The edger marks every tested lip cell; the radius-section maximum error is 1.12 mm. These are short component proofs, not a whole pad or four-side completion.

6.5 Local joint finishing, troweling and cleanup

The T-junction fixture uses three pitched head strokes, flat head, bed and corrective head paths. A 20 N and a matched 2 N run use the consolidation law of section 3.11 at the same 37-minute condition. Peak finite carriage forces are 32.22 and 6.44 N. The positive run transfers 1.324 ml in five disclosed **phantom roll**

wipes to the mortar board; the light control transfers 1.361 ml. Joint cells are never assigned a finishing state. Volume errors are at floating-point roundoff; no simulation fault or runtime warning was recorded.

Case	Fixed ROI	Cells	Concave	Median C	C >=0.5
joint	head	45	95.6%	0.710	80.0%
joint	bed	280	100.0%	0.651	73.9%
joint	cross	2	100.0%	0.679	100.0%
light	head	45	75.6%	0.259	0.0%
light	bed	280	0.0%	0.151	0.0%
light	cross	2	0.0%	0.303	0.0%
cleanup	head	45	95.6%	0.710	80.0%
cleanup	bed	280	100.0%	0.651	73.9%
cleanup	cross	2	100.0%	0.679	100.0%

The 20 N case establishes a positive compacted local core and concavity; endpoint/edge cells leave only 80% head and 74% bed above C=0.5, so this is **not** full-coverage joint completion. The light control reaches zero qualifying C coverage. The two crossing cells are a narrow profile check, not a large-area statistic. Entire-infill task criteria remain incomplete.

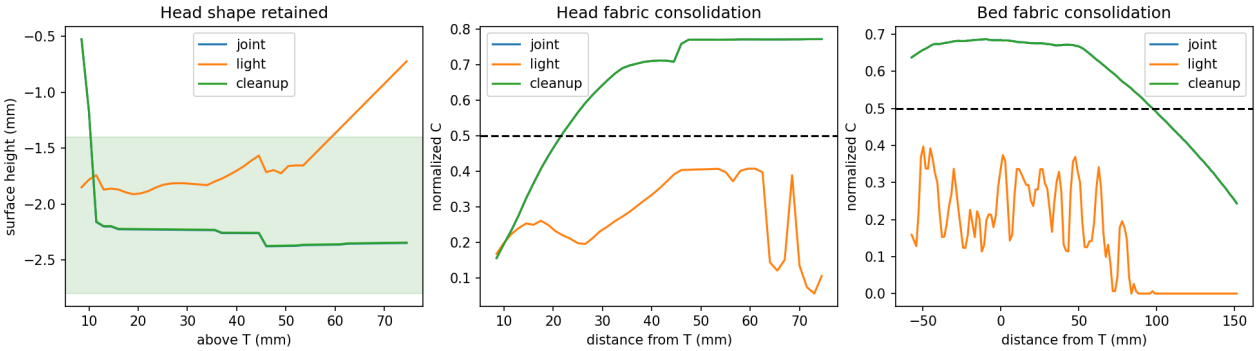


Figure 12. Native pressure-dependent fabric consolidation with head/bed profiles. Cleanup preserves the useful local joint finish.

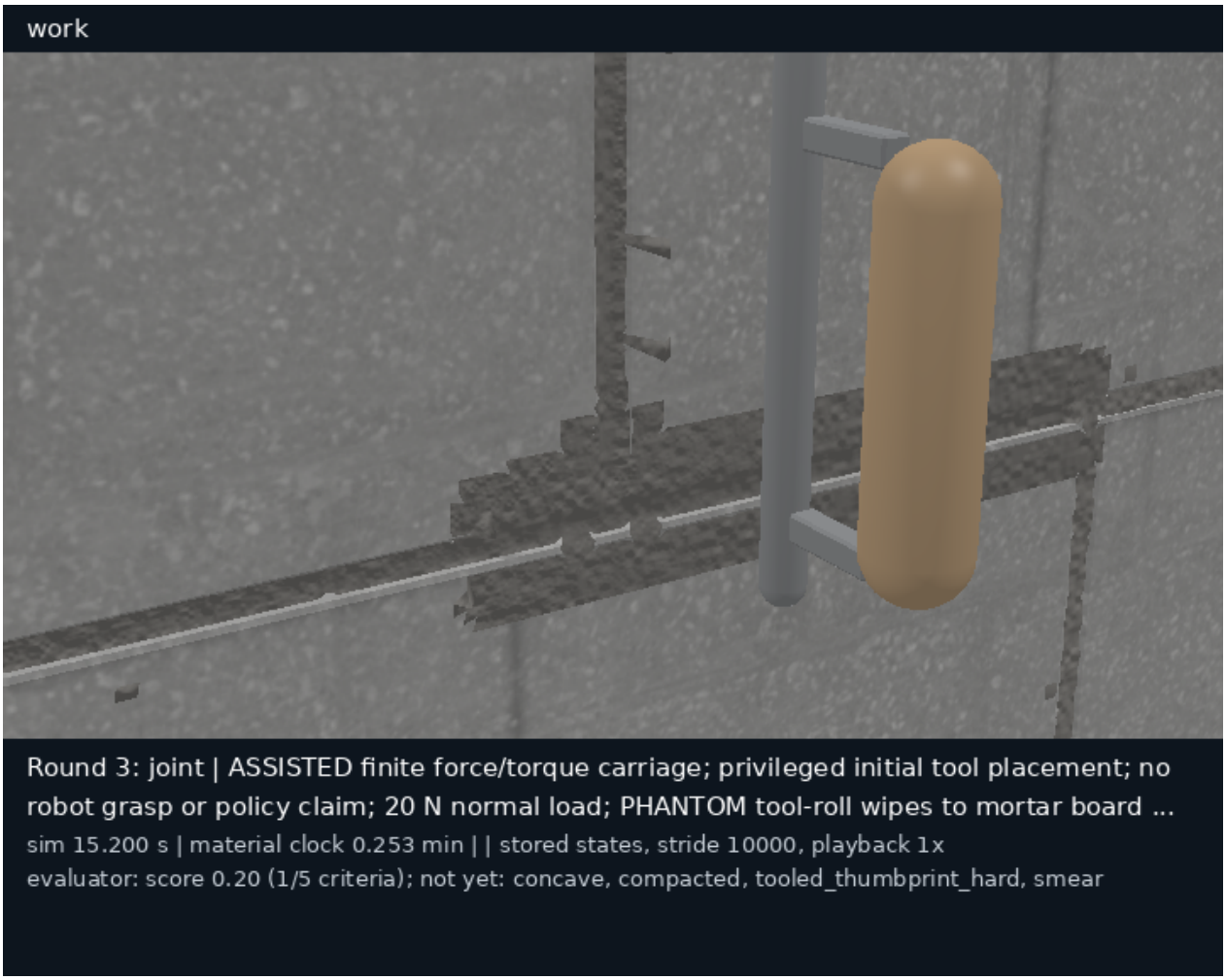


Figure 13. Oblique exact final joint state after withdrawal. Residual sheets/tags are deliberately visible; concavity does not imply clean block faces.

The steel-trowel sample uses the same authored already-floated start and six native 20 N, 2-degree passes. In the fixed 100x200 mm central region, all 5,000 cells satisfy D >=0.8 and a <=0.05 mm: texture min/median/max 0.0415/0.0451/0.0477 mm and D about 0.99999. Height range is numerically flat. Excess at stroke ends reaches 24.43 mm; this is a local smooth-hard finish, not clean whole-pad completion.

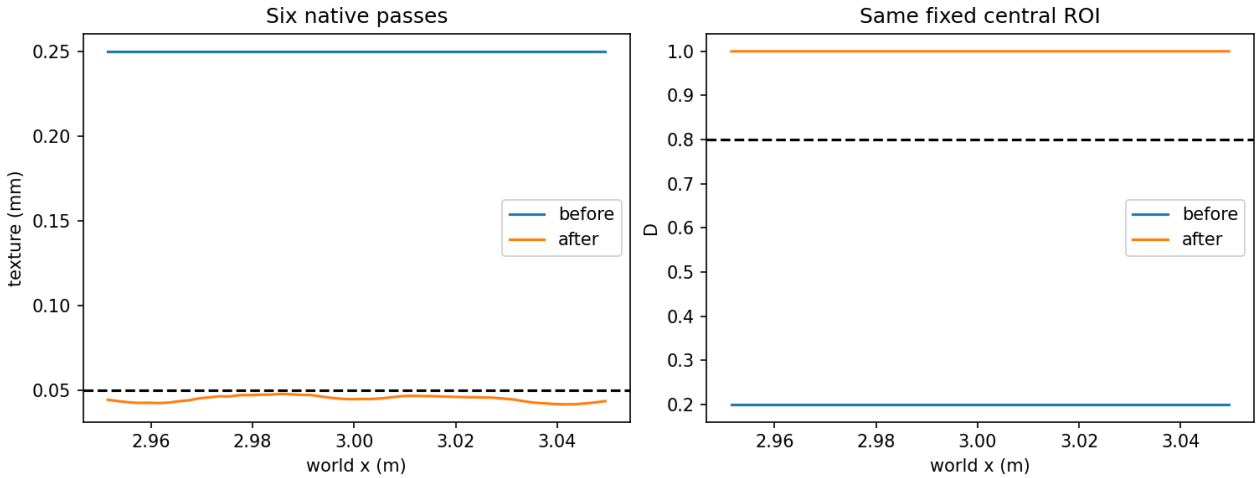


Figure 14. Combined trowel finish from the same saved physical record, with unchanged texture/density thresholds.

A separate margin-trowel fixture starts from the actual saved positive joint material arrays and makes a

native 6 N scrape above the bed. It removes 96.95% of a selected 0.935 ml mortar tag, leaves 0.0285 ml in that face region, and carries 0.864 ml. The remainder is visible nearby displacement, not deletion. Maximum centre-line joint change is below 0.008 mm, and the concavity and consolidation shares above survive. No phantom wipe is used in this cleanup. This qualifies one same-material tag, not disposal of all tags or the old repair-mortar pile.

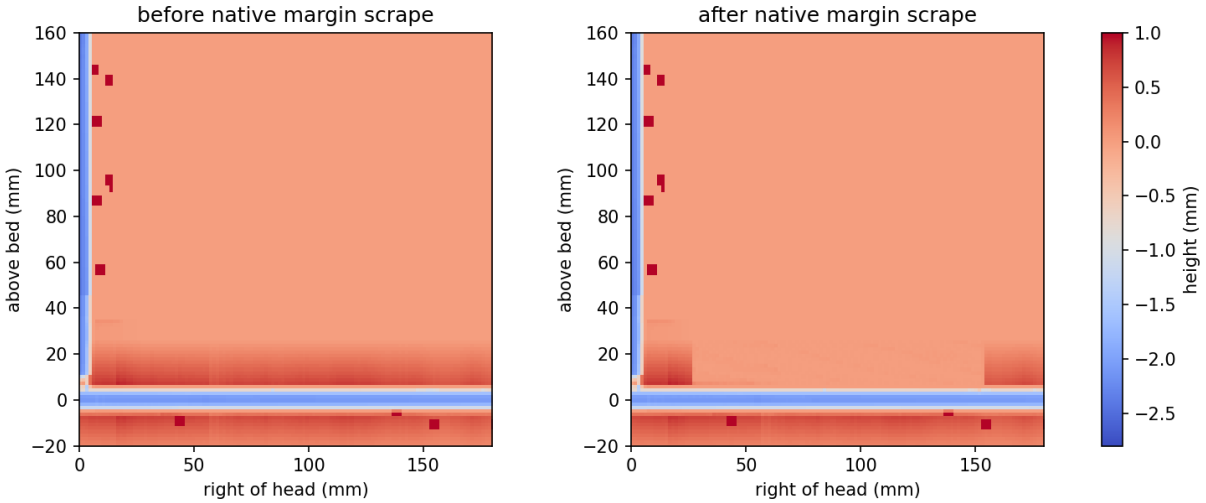


Figure 15. Spatial before/after height of native mortar-tag recovery; material goes to the lifted margin trowel.

The four films of these runs contain 10 Hz exact-state replays and disclose assistance. Sensitivity checks call native material work on prescribed fields, separately from those rigid trials. All tested concrete-rate alternatives preserve the direction of smoothing/embedding/densification and reach smooth-hard after 12 full-contact constitutive passes. Exact pass count is sensitive. Mortar C at 20 kPa/150 mm slip ranges 0.639–0.985 for individual uncertainty ends; a paired thick/slow case yields 0.488, requiring more work. The model does not claim identical success under every plausible parameter choice.

7 Runtime and entry points

Routine policy RGB uses Mesa software OpenGL, reading the same exported USD robot/room meshes, diffuse textures and exact dynamic paste geometry. It uses filtered physical texture-amplitude shading and broad directional/ambient/specular lighting; it does not reproduce RTX shadows, reflections or all MDL detail. Recorded presentations can select Isaac explicitly; Isaac camera sessions exceeded 200 times real time and are not part of the measured routine path.

Configuration	Count	Launch s	Max wall/rigid time
pad-default	12	32.76	16.78
three-cameras	15	27.29	15.87
hd-single	12	25.54	16.22
assisted contact + recording	6	24.19	45.92

Startup and paused-worker restart are reported separately from action ratios. Image download is included. No material-wait time or artificial minimum-action padding is added to the denominator: the shortest command remains 0.05 rigid seconds. The contact workload uses the normal HTTP facade/backend/codec with an explicitly privileged float fixture; it is not a managed CLI launch or a policy solution. The three normal camera sessions are actual `gar run` launches. Both robot configurations execute real actions.

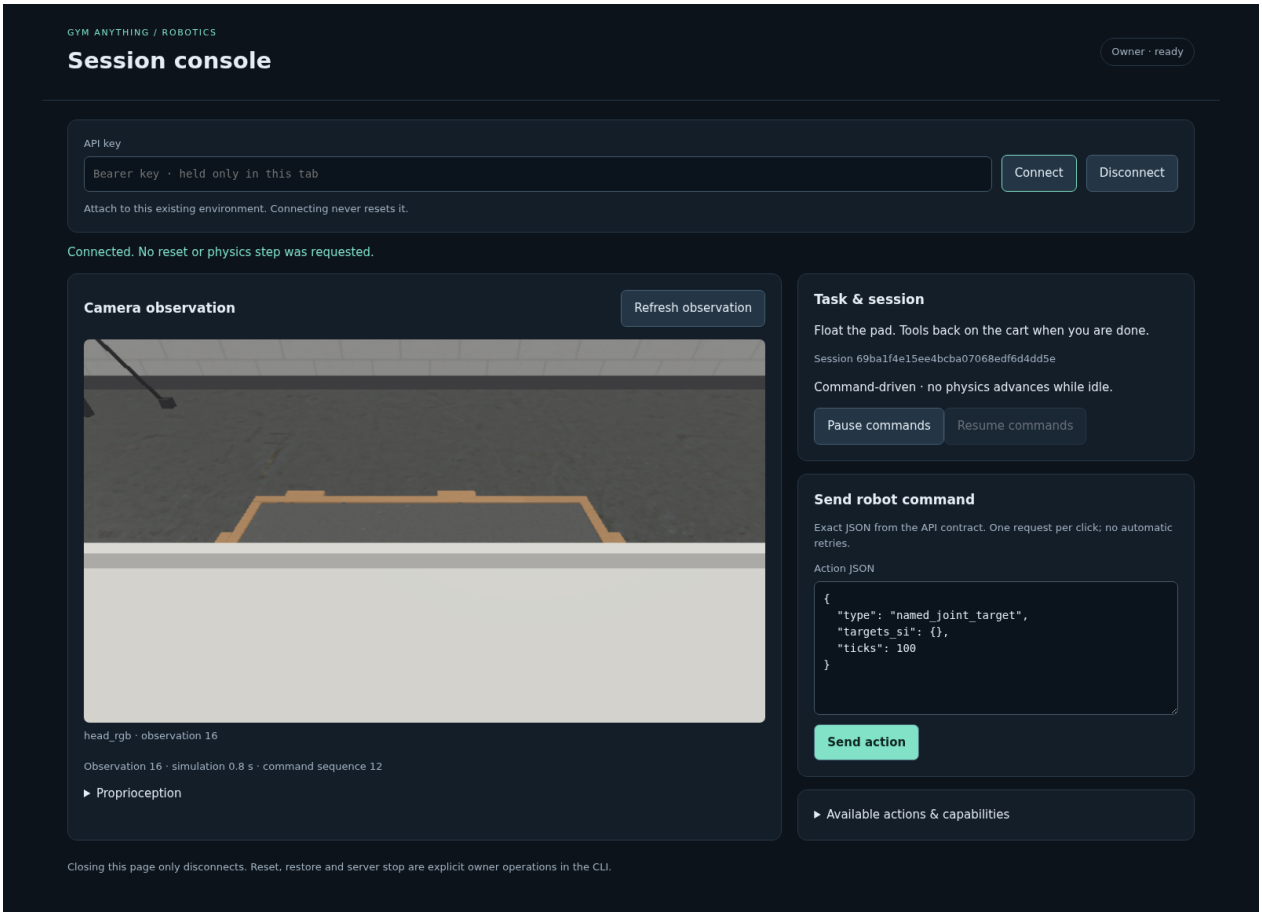


Figure 16. The owner's dashboard after legal commands, with task, camera and controls. The restricted dashboard was also exercised; credentials are not stored.

Regression checks: 29 passed, three optional full-world tests skipped, three subtests passed. Separate direct checks cover window ownership, support, grid shifts, floor catches, restoration, source history and fibre-top coverage. Live camera sessions exercise construction, legal commands, camera delivery, pause/restart and shutdown. The discovery, inspect, tasks and doctor commands pass.

MuJoCo Warp emits a construction warning about some shape pairs lacking multicontact support. This world uses MuJoCo CPU stepping and contact, with multicontact enabled; the warning comes from Warp's

model mapping. The component records report no simulation fault or captured solver warning. This does not make every initialization warning irrelevant.

8 Reproduce and inspect

The environment is the package `concrete_finishing`, started and driven with the `gar` command. Discovery and task help:

```
gar list
gar inspect concrete-finishing
gar tasks inspect concrete-finishing
gar doctor concrete-finishing
gar run concrete-finishing --select task=pad-float --select seed=0 --policy-key \
  --format json
gar run concrete-finishing --robot mobile-fr3-single --select task=tool-sealant \
  --policy-key --format json
```

A recording holds exact state arrays and portable scene assets, not native simulator checkpoints. A film is rendered from a recording without repeating the physical actions, and replay checks the ordered ticks before rendering.

9 Limitations

- The finite reserve stops continuing weak-layer regeneration, but bearing/squeeze and transport remain reduced-order approximations. Texture/densification, spill/roll geometry, 20 mm overflow, adhesion and skin parameters need broader sensitivity/calibration.
- Native local troweling reaches smooth-hard in its fixed central sample; mortar C reaches a positive core but misses full coverage at endpoints. Large excess piles and many joint tags remain. The native cleanup qualifies one mortar tag, not the repair-mortar pile or a clean wall. Model coefficients remain estimates with explicit sensitivity, not physical calibration.
- Mesh coverage and graded feathering are demonstrated with explicit assistance. The solid-volume/contact-envelope distinction is physical, but the homogenized weave and one-dimensional blade compliance omit distributed flexure.
- Catch spills are located, visible and recoverable. The two seam scrapes recover their small deposits; other partial cleanup leaves a real remaining penalty. Intertile flux, mixed species and arbitrary dropped-blob collisions are outside the model; no irreversible invisible counter is used for new deposits.
- Skin timing uses actual material age, and late tooling has consequences. The factor-three surface resistance and torn-skin rule are estimates; no polymer chemistry or measured fracture curve is claimed.
- Active paste contact covers nominated soles/fingertips, not the full robot or every orientation of a dropped tool. The penetrometer has no moving readout ring. The gripper holds weakly against torsion, as robot attempts showed.
- Only mobile dual-FR3 and mobile single-FR3 are supported. No humanoid qualification is claimed. Software RGB is geometrically faithful with simplified lighting; optional Isaac remains outside the measured runtime qualification.

10 Sources

Opened (read on the page or in the PDF):

- NRMCA *Concrete in Practice*: CIP 1 "Dusting Concrete Surfaces", CIP 13 "Concrete Blisters", CIP 14 "Finishing Concrete Flatwork", CIP 20 "Delamination of Troweled Concrete Surfaces".
- FHWA-RD-02-099, *Guide for Curing of Portland Cement Concrete Pavements* (2005): the evaporation equation of ACI 308.
- Ferraris and de Larrard, NISTIR 6094, *Testing and Modeling of Fresh Concrete Rheology* (1998), eq. 19 (slump to yield stress).
- Roussel and Coussot 2005, *J. Rheol.* 49, 705 (the spread-regime relation; formula derived from Liu, Balmforth and Hormozi 2018, *J. Non-Newtonian Fluid Mech.* 258, eq. 9).
- Clayton, Grice and Boger 2003, *Int. J. Mineral Processing* 70 (Pashias et al. 1996's cylinder slump model).
- Yu, Chandra, Wilkes, Zhao and Soga 2025, arXiv:2512.23364 (Roussel 2006's thixotropy classes reproduced).
- Humboldt H-4134 and H-4137 penetrometer manuals (ASTM C403 set definitions; the H-4134 needle of 1/20 in^2).
- MASCO *Concrete Forming & Accessories* catalogue (2016), the setting-time table.
- Femmam, Mabrouki and co-authors 2018, *J. Appl. Eng. Sci. Technol.* 4(2), Table 1 (Green 1954's inclined-load bearing factor).
- Le Thanh Hai and Phan Van Tien 2014, *Journal of Science and Technology* (Vietnam) 52(6) (probe tack of fresh paste).
- Kobayashi, Soma and Shigihara, 13th ISARC 1996 (a robotic trowel's pressing forces and speeds).
- Brick Industry Association Technical Notes 7B (2017) and 10 (2009).
- Sika Canada Sikaflex-1A and Tremco Dymonic 100 data sheets; SikaQuick Pro 1000 and Quikrete FastSet Repair Mortar data sheets; 3M US 11,267,758 B2 (wall compound yield stress and viscosity; read through a text extraction).
- NIH Office of Research Facilities, *News to Use*, "Sealant Joints – Part 2: Joint Design" (March 2021): coplanar joints "tooled flush to slightly concave".
- Chemical Concepts' Albion caulking spatula pages: tooling spatula widths 3/8 to 1-1/2 in; "choose the width that matches the joint".
- Marshalltown product pages for every tool modelled.
- U.S. Access Board, *Guide to the ADA Standards*, chapter 3 (§303 changes in level: the patch's lip criterion).
- Repository: `robots/models/mobile-fr3-duo`, `classroom_arrangement` (the robot, its controllers and reflexes), `sweeping` (the GAR backend's render-worker lifecycle and the report converter), `drilling` (render stage, dressing services, white balance), `ingredient_mixing` (render worker), Poly Haven assets as pinned in `concrete_finishing/render.py`.

Estimates are labelled in the ledger (section 5) with their reasons; snippet-only facts (search summaries) were not used as values without the label.